Response Letter with Annotations

Robertson, R. E., Green, J., Ruck, D. J., Ognyanova, K., Wilson, C., & Lazer, D. (2023). Users choose to engage with more partisan news than they are exposed to on Google Search. *Nature*, *618*(7964), 342-348.

Annotations

1. **Purposes**
    - **a.** Motivation (one-sentence summary)
        - This study is motivated by the question of whether partisan news consumption reflects users' choices or the content prioritized by search algorithms.
    - **b.** Specific Rationale (several bullet points)
        - Much of the literature on selective exposure and partisan media consumption relies on **observed engagement behaviors** (e.g., clicks or visits), which confound algorithmic curation with user choice.
        - On platforms like **Google Search**, users are first exposed to a ranked set of results, but **exposure does not guarantee engagement**; therefore, observing only what users click may overstate the role of algorithms or misattribute user agency.
        - Existing research has limited direct evidence comparing:
            1. what users are **shown** (exposure),
            2. what they **choose after being shown** (follows),
            3. and what they **consume overall** (engagement).
        - By combining **survey data** with **observational digital trace data**, the authors can link political characteristics (e.g., partisanship) to real-world search behavior.
    - **c.** Research Questions and/or Hypotheses
        - Main research question:
            1. Do users choose to engage with more partisan news than they are exposed to on Google Search?
                - **a.** How does the partisanship of exposure differ from the partisanship of engagement?
                - **b.** Does this gap between exposure and engagement vary by users' political partisanship?
                - **c.** How does engagement that directly follows exposure ("follows") compared to overall engagement?
        - Identify IV, DV, mediator, moderator
            1. DV
                - **a.** Partisanship of news engagement
                    - **i.** Measured as the average partisan bias of news domains users visit.
                - **b.** Engagement with unreliable news domains
                    - **i.** Count of visits to unreliable news domains.
            2. IV
                - **a.** Type of interaction with content

i. Exposure
ii. Follows
iii. Overall engagement
b. User political partisanship

- Identify specific relationships among variables
    1. **1.** Engagement with news content is more partisan than algorithmic exposure via search results.
    2. **2.** Engagement patterns vary by partisan identity, with strong partisans exhibiting greater ideological selectivity.
    3. **3.** Engagement patterns vary by partisan identity, with strong partisans exhibiting greater ideological selectivity.
    4. **4.** Engagement conditional on exposure ("follows") falls between exposure and overall engagement in terms of partisan intensity.

2. Method

a. Research Design

- Survey
    1. Repeated cross-sectional surveys conducted in two waves.
    2. Survey data provide individual-level characteristics (e.g., partisanship, demographics) that are linked to behavioral data via participant IDs.
- Observational data
    1. Passive observational data collected via a custom browser.
    2. Tracks real-world Google Search behavior.
        a. search result pages
        b. website visits
        c. follows

b. Sample (representative sample?)

- Participants: U.S.-based internet users who voluntarily installed the browser extension.
- Sample size varies by wave
    1. 2018 wave
        a. YouGov recruitment
        b. In 2018, the authors collected exposure to 102,114 Google Search result pages for 275 participants, follows on 279,680 Google Search results for 262 participants and overall engagement with 14,677,297 URLs for 333 participants.
    2. 2020 wave
        a. PureSpectrum quota sample
        b. In 2020, the authors collected exposure to 226,035 Google Search result pages for 459 participants, follows on 69,023 Google Search results for 418 participants and overall engagement with 31,202,830 URLs for 688 participants.
- Representativeness

1. The sample is not nationally representative. Participants were recruited through online panels and voluntarily installed a browser extension, with oversampling of strong partisans in the 2018 wave and quota sampling in 2020.

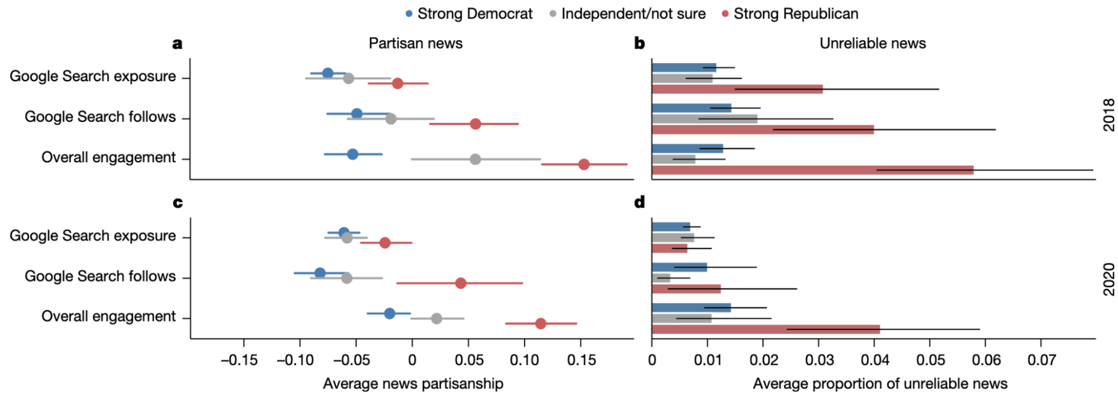
c. Measures
- Survey
  1. Partisan identification (7-point scale)
  2. Age
- Observational data
  1. Google Search Exposure
     a. URLs shown on the first page of Google Search results.
  2. Overall engagement
     a. URLs visited during browsing activity.
  3. Google Search Follows
     a. Visits to URLs that appeared in search results within a short time window.

d. Analysis
- Analyses include descriptive comparisons, nonparametric tests (e.g., Kruskal–Wallis), and regression models (OLS and negative binomial) to examine differences between exposure and engagement.

3. Results and Conclusions

a. Answers to RQs and Hypotheses
- Do users choose to engage with more partisan news than they are exposed to?
  1. Yes. Engagement (both within Google Search and overall web use) is more identity-congruent and more unreliable than the news shown in Google Search results.
- Exposure vs. engagement partisanship
  1. Partisan differences are small for exposure, larger for follows, and largest for overall engagement.
- Variation by users' political partisanship
  1. Partisan identification has only a small, inconsistent relationship with exposure, but a stronger and more consistent relationship with follows and overall engagement, meaning stronger partisans select more identity-congruent and unreliable sources when they choose what to click.
- Follows vs. overall engagement
  1. "Follows" (clicks directly after seeing a result) are already more partisan than exposure, but overall engagement is even more partisan and unreliable.

b. Include Key Figures/Tables



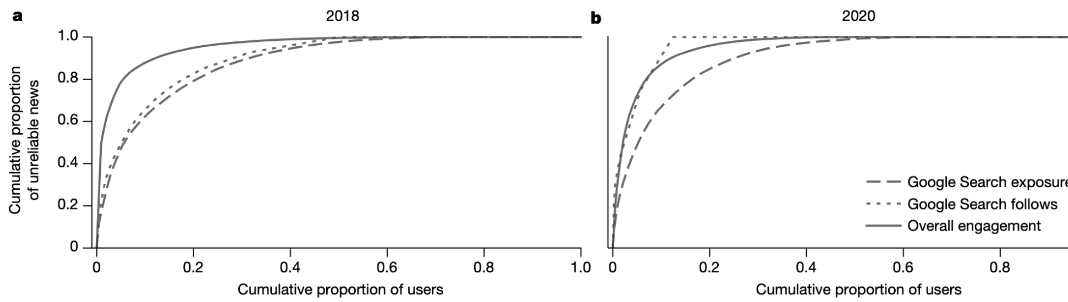
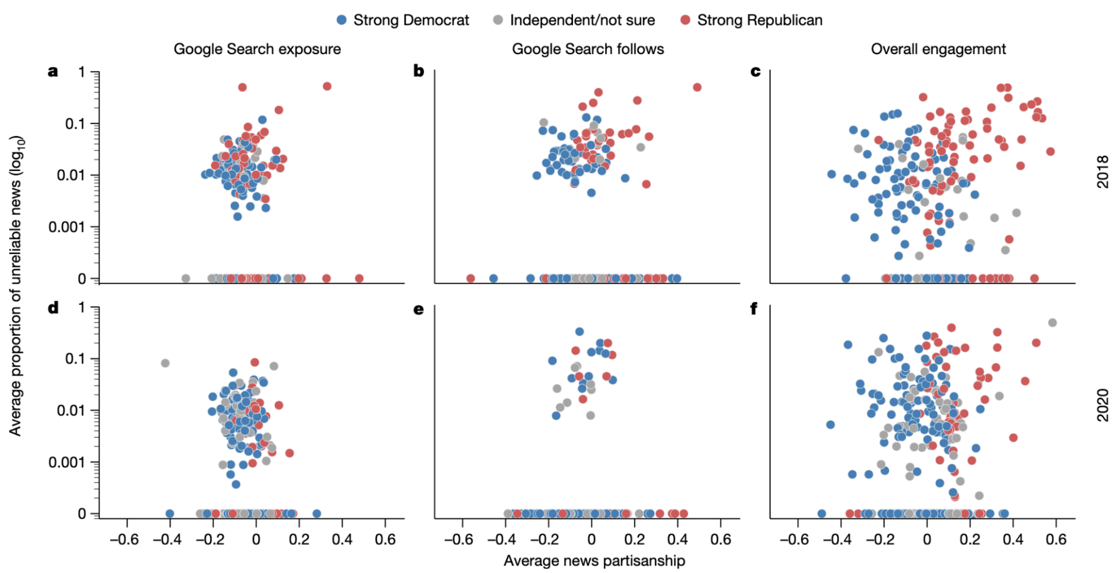
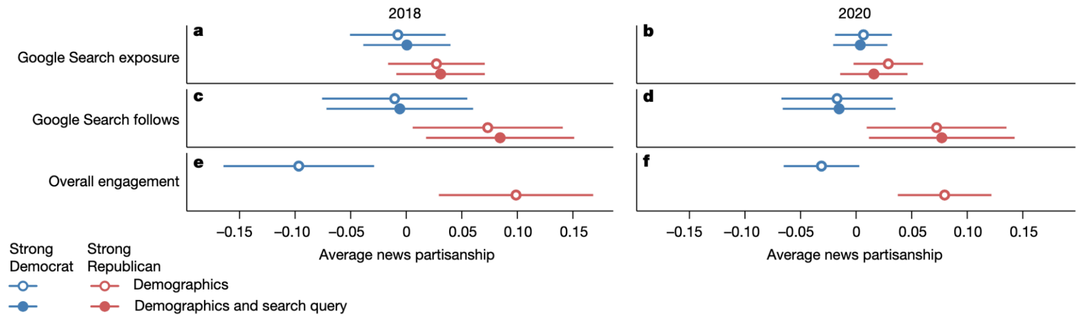


Fig. 3 | Exposure to unreliable news is less concentrated among a small number of participants than follows or overall engagement. a,**b**, Each line shows the cumulative proportion of participants (x-axis) that account for the cumulative proportion of unreliable news seen by all participants (y-axis) within each data type (legend) in 2018 (**a**) and 2020 (**b**).





c. Authors' Conclusions

- User choice, not Google's algorithm, is the primary driver of partisan and unreliable news engagement in this setting.
- Google Search results themselves show relatively limited partisan skew, and the platform tends to increase ideological audience diversity.
- **Concerns about search-driven "filter bubbles" are not strongly supported here**; instead, selective engagement by users amplifies partisanship and unreliability beyond what they are exposed to.

4. Discussion

a. Implications

- While user choice is the primary driver, Google's algorithms are not necessarily normatively unproblematic, as even limited exposure to unreliable news can have negative impacts.
- Determining when to show or omit partisan content is an evolving and adversarial challenge requiring deep social context and expertise.
- The study's method of collecting data directly from users offers a viable way to assess algorithmic accountability when industry data-sharing is limited.

b. Limitations

- Sample Representation: The study relied on relatively small samples that were not fully representative of the general US population.
- Domain-Level Metrics: Using domain-level scores limits the ability to detect partisan or reliability differences between individual articles within the same website.
- Device and Platform Scope: Data were only collected from desktop computers and focused exclusively on Google Search, excluding mobile traffic and other search engines.
- Search Depth: The study only analyzed the first page of search results (SERPs), missing context from results beyond the first page.

c. Future Directions

- Expanded Scope: Future research should examine mobile devices, other search engines (e.g., Bing, Yahoo), and search behavior in non-US or non-English contexts.

- Diverse Topics: Investigating other socially important topics beyond the 2018 and 2020 US elections is necessary.
- Emerging Technologies: Research is needed to explore how new technologies, such as AI chatbots or messaging apps, influence user exposure and engagement patterns.

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Response Letter (~600 words)

- Your assessment of the study
    - Overall, this is a really solid and inspiring piece. First of all, I am particularly interested in the **“Exposure-Engagement”** Framework. The study’s most significant contribution is the systematic comparison between exposure (what the algorithm shows) and engagement (what the user chooses). This allows the authors to disentangle the effects of algorithmic curation from human agency, providing a clear empirical answer to the “filter bubble” vs. “echo chamber” debate.
    - Moreover, the **immense workload** and methodological integrity of this research are further validated by its impressive longitudinal scope. Rather than relying on a one-time observation, the study spans two highly complex U.S. election cycles in 2018 and 2020. Despite the different social and political climates of these two periods, the findings remain remarkably consistent, repeatedly pointing to user choice as the primary driver of polarization. The ability to replicate such empirical results across two major political milestones speaks volumes about the depth of the project's engineering and the robustness of its logical framework. It is, without a doubt, a top-tier piece of work that pushes empirical social science to its absolute limit.
    - I think one of the most brilliant parts of this research is how it handles **the black box problem of Google’s algorithm.** Usually, it is almost impossible for us to study these platforms because their internal algorithms are top secrets. But instead of trying to get data from Google, the authors were very clever, they went directly to the users. By using a browser extension to record exactly what appeared on real people’s screens, they managed to see the output of the algorithm without needing to see the code itself. This user-side approach is like a textbook example for us: it shows that even if tech companies do not share their data, we can still find a way to independently audit their impact from the outside. It is a very smart way to bypass the platform’s restrictions.
- Weaknesses
    - A critical limitation of this study is the potential lack of **generalizability to platforms** with different algorithms, specifically social media. The findings here are contextualized within the **“Pull-based”** logic of Google Search, where algorithms primarily respond to explicit user intent (queries) by prioritizing relevance and authority. However, the dynamics of partisan amplification may be fundamentally different on **“Push-based”** platforms like Twitter (X), TikTok, or Reddit. On these platforms, the Social Network (Social Graph) plays a decisive role:
        - Engagement-Driven Push: Unlike Google, social media algorithms are often optimized for maximum retention and virality. They actively “push” content based on a user’s social connections and past emotional engagement, which may create a stronger algorithmic "filter bubble" than a search engine does.

        - The “Social Echo Effect”: While this study finds that user choice is the primary driver in search, on social media, the line between user choice and algorithmic curation is blurred. A user’s choice to follow a partisan friend (social choice) triggers an algorithm to show more similar content (algorithmic choice), creating a self-reinforcing feedback loop that may not exist in the more linear search-and-click process of Google.
    - Consequently, while this research effectively deconstructs the “filter bubble” in the context of web search, further research is required to determine if the same user-choice-dominant model holds true in environments where social-algorithmic curation is the primary mode of content consumption.
- Other notes